Mechanisms · Knowledge Organiser

Learning goal: I can identify levers, pulleys, and gears and explain how they let a small effort move a heavy load.

Today I Can

I can label the pivot, effort, and load on a simple lever diagram.

I can match everyday tools to levers, pulleys, or gears.

I can explain that a mechanism trades a longer push distance for a smaller effort force.

Must-Know Words

mechanism: a simple machine such as a lever, pulley, or gear that changes a force

lever: a stiff bar that turns on a pivot to lift or move a load

pivot: the fixed resting point (fulcrum) that a lever turns around

pulley: a grooved wheel with a rope that changes the direction or size of a pull

Core Science Steps

First, look at the lever: it has a load, a pivot, and an effort push.

When the effort push is far from the pivot and the load is close, a small push lifts a big load.

Pulleys use a wheel and rope, and gears use interlocking teeth to change force, speed, or direction.

Keep the Science Clear

Identify the effort, pivot (or wheel/teeth), and load before explaining how the force changes.

Diagrams show ideal machines; real levers, pulleys, and gears also have some friction at the axle or pivot.